

QUANTUM LDPC CODES

A Comprehensive Project Report

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1. Introduction:

The goal of this project is to understand Quantum Low-Density Parity-Check (LDPC) Codes and their application in quantum error correction. The focus is on understanding how LDPC codes handle bit-flip and phase-flip errors in qubits. The project further aims to simulate the encoding and decoding of quantum LDPC codes and analyze their performance to draw meaningful inferences.

2. Qubits and Basic errors:

A qubit is the fundamental unit of quantum information. Unlike a classical bit which is either 0 or 1, we represent qubits using computational basis states $|0\rangle$ and $|1\rangle$ also known as the Z-basis, where $|0\rangle = [1 \ 0]^T$ and $|1\rangle = [0 \ 1]^T$. In general a quantum state is represented as a superposition of computational basis $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex probability amplitudes. The squares of their magnitudes gives the probability of measuring $|0\rangle$ or $|1\rangle$ respectively, and they satisfy the condition $|\alpha|^2 + |\beta|^2 = 1$.

The Bloch sphere provides a useful visual representation of a qubit state, while the vector representation is used for mathematical computations.

The errors occur in a quantum system mainly due to noise from the surroundings. There are mainly two errors that occur which are bit flip error and phase flip error. The Pauli X gate corresponds to bit flip which swaps $|0\rangle$ and $|1\rangle$ detected via z basis measurement and Pauli Z gate corresponds to phase flip error and detected via x basis measurements.

Multiple qubits are represented as superpositions of all possible computational basis states by the n qubits. Ex:- 2 qubits can be written as $|\phi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$.

3. Quantum measurements:

Quantum measurements are described by a collection $\{M_m\}$ of measurement operators. m refers to the measurement outcomes that may occur in the experiment. If $|\psi\rangle$ is the state of the qubit before measurement, probability that result m occurs is $p(m) = \langle \psi | M_m^\dagger M_m | \psi \rangle$. The state of the system after measurement is $\frac{M_m |\psi\rangle}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}}$. Projective measurement is when the measurement operators are orthogonal projectors onto basis states. For example, measurement operators in the Z-basis are the projectors $P_0 = |0\rangle\langle 0|$ and $P_1 = |1\rangle\langle 1|$. Similarly, measurement in the $|+\rangle, |-\rangle$ basis corresponds to projection onto eigenstates of the Pauli-X operator.

4. Classical LDPC codes:

Before understanding quantum LDPC codes, classical LDPC codes were studied as a foundation to understand more. A classical LDPC code is defined by a sparse parity-check matrix H , and valid codewords c satisfy $Hc^T = 0 \pmod{2}$. The sparsity of H enables efficient iterative decoding algorithms. The generator matrix G was obtained by computing the null space of H , allowing encoding through $c = mG \pmod{2}$.

Decoding was implemented using the belief propagation algorithm on the Tanner graph representation of the code. In this representation, variable nodes correspond to codeword bits and check nodes correspond to parity constraints. Log-likelihood ratios (LLRs) were initialized based on the channel crossover probability p , with prior reliability $L_0 = \log\left(\frac{1-p}{p}\right)$. If a bit $c_i = 1$ then $LLR[i] = -L_0$, and if $c_i = 0$ then $LLR[i] = +L_0$. Messages were then iteratively passed between variable and check nodes.

Variable-to-check messages are computed as $m_{v \rightarrow c}^{(t)} = L_v + \sum_{c' \in N(v) \setminus c} m_{c' \rightarrow v}^{(t-1)}$ and check-to-variable messages as $m_{c \rightarrow v}^{(t)} = 2 \tanh^{-1} \left(\prod_{v' \in N(c) \setminus v} \tanh \left(\frac{m_{v' \rightarrow c}^{(t-1)}}{2} \right) \right)$. After each iteration a tentative decoded codeword was formed and verified using the parity-check condition $H \cdot \text{decoded} \equiv 0 \pmod{2}$. The process terminated either when a valid codeword was found or when the maximum number of iterations was reached.

A complete simulation was developed to generate random messages, encode them, transmit them through a noisy channel, and attempt recovery using belief propagation. This allowed observation of decoding behavior and error-correction performance under varying noise probabilities. The necessary corrections to the classical BP decoder were subsequently made and graphs were added to compare performance at varying noise probabilities and maximum iterations.

4. Quantum error correction and stabilizer code:

Then we studied the paper "Degenerate Quantum LDPC Codes With Good Finite Length Performance" by Pavel Panteleev and Gleb Kalachev. Errors are represented by Pauli operators in case of single qubit it can be X (bit flip), Z (phase flip), Y (both bit and phase flip) and I (no error). But for a multi qubit system an error is a tensor product of these Pauli matrices.

To detect errors without directly measuring the data qubits, the stabilizer is used. A stabilizer is a tensor product of Pauli operators. To protect a multi qubit system we introduced the concept of stabilizer code defined as subspace where the state $|\psi\rangle$ is a $+1$ eigenstate of a commutative subgroup of Pauli operators, expressed as $M|\psi\rangle = +|\psi\rangle$. Instead of measuring each qubit the stabilizer measures the parity of the qubits. If a physical error occurs that anti-commutes with our stabilizer, it forces a negative eigen value indicating an error occurred without disturbing the logical encoded data. Each stabilizer must be linearly independent and pairwise commute. An error is detected based on measurement of binary vector syndrome s .

5. How syndrome is measured:

The direct measurement of data qubits collapses the encoded quantum information. We use an auxiliary ancilla bit. First the ancilla qubit is prepared in a superposition state by applying a Hadamard gate, transforming $|0\rangle_A$ into $\frac{1}{\sqrt{2}}(|0\rangle_A + |1\rangle_A)$. Controlled Pauli gates are then applied between the ancilla and data qubits, which allows parity information of the stabilizer to become entangled with the ancilla. Again a Hadamard gate is applied to create quantum interference and an ancilla qubit is finally measured to obtain classical syndrome bits.

5. Binary symplectic representation:

In order to use classical decoding algorithms we represent quantum operators in binary form. Pauli operators are mapped into a binary array of the form $(x | z)$ as follows: $X \rightarrow (1|0)$, $Y \rightarrow (1|1)$, $Z \rightarrow (0|1)$, $I \rightarrow (0|0)$. To check if an error and a stabilizer commute, the symplectic inner product is computed classically as $\langle x_1, z_2 \rangle + \langle z_1, x_2 \rangle \pmod{2}$. A result of 0 means they commute and no error is detected, while a result of 1 means they anti-commute and the syndrome alarm triggers.

The parity matrix is represented by $H=(H_x|H_z)$ where H_x represents the bit-flips and H_z represents the phase-flips in each qubit. For the code to be valid, all the stabilizers must commute, i.e., $H_x H_z^T + H_z H_x^T = 0$. By defining $H_x=[A,B]$ and $H_z=[B^T, A^T]$ where A and B are circulant matrices, commutativity ($H_x H_z^T = AB + BA = 0$) is guaranteed due to the properties of the ring of circulants. The stabilizer generators form the rows of the global check matrix $H = [H_x | H_z]$, and the complete syndrome vector is computed as $s = (H_x e_z^T + H_z e_x^T) \pmod{2}$. Once H is constructed, we solve for $He=s$.

6. Syndrome based BP decoder:

The quantum BP decoder differs critically from the classical one in that it operates exclusively on the measured syndrome rather than on the received codeword. Variable nodes represent qubit errors and check nodes represent stabilizer measurements. All LLRs are initialized to $+L_0$, assuming no error by default. The syndrome is hardcoded into the check nodes and is used to determine the parity of errors among connected qubits. The check-to-variable messages incorporate the syndrome by flipping the sign of the product when $s[i] = 1$. Messages are iteratively passed between check and variable nodes, gradually estimating an error vector consistent with the measured syndrome. The convergence check verifies $He = s \pmod{2}$. If not satisfied, iteration continues until the maximum iteration count is reached, at which point the best available estimate is returned.

7. Why BP fails on Quantum codes and solution is OSD:

Even though BP works very well in classical codes it often fails quantum LDPC codes due to two reasons. First one due to the short cycle in the Tanner graph which causes correlated messages and prevents convergence. Another is due to degenerate errors, where multiple physical errors produce the same syndrome which BP fails to distinguish them.

To solve this we use Ordered Statistic Decoding (OSD) as a post process step. It uses the soft beliefs (LLRs) outputted by BP. The columns of the parity matrix H and the estimated error vector e are sorted based on the magnitude of these LLR values, where the most reliable bits have the largest absolute values. A residual syndrome is computed in order to find the remaining error after BP decoding. A set of linearly independent columns corresponding to the most reliable bits (the Information Set) is selected and solved to obtain the required correction. Finally, the solution is permuted to get the corrected vector back into its original order.

6.Implementation and results:

We implemented the BP+OSD algorithm for quantum LDPC codes based on the stabilizer code framework, where Pauli errors are represented using binary symplectic representation and decoding is formulated as solving linear systems. The Word Error Rate(WER) was plotted against Physical error rate. We plotted the error curves for both BP alone and BP+OSD decoders and compared. The OSD post processing step significantly improved the efficiency of the decoder. Which can be interpreted by the lower WER. The gap between the two curves indicates the error that BP alone could not resolve but OSD successfully corrected using reliably sorted Gaussian elimination.

7.Conclusion:

This project demonstrated a complete pathway from understanding the fundamental properties of qubit to a working quantum error correction decoder. By representing fragile quantum states and Pauli operators in binary symplectic representation, which makes it possible to apply classical decoding techniques to quantum error correction. The results further showed BP alone often fails on quantum LDPC codes, due to short cycles in Tanner graph and degenerate errors. However, the syndrome based BP decoding combined with reliability-sorted OSD correction yields a robust quantum decoder. By bridging quantum hardware with efficient classical algorithms through the binary symplectic representation, quantum LDPC codes provide a scalable path toward fault-tolerant quantum computing.